SECTION 1: ELECTRIC CURRENT & CHARGE FLOW (Q1–20)

1. Define electric current. What is its SI unit?

- **Definition:** Electric current is the rate of flow of electric charge through a conductor.
- **Formula:** $I = \frac{Q}{t}$, where Q is charge, t is time.
- **SI unit:** Ampere (A), where 1 A = 1 coulomb/second.

2. State the relationship between current, charge, and time.

$$I = \frac{Q}{t}$$

- I = current (A)
- Q = charge (C)
- t = time (s)

This means **current is the charge flowing per unit time**.

3. How many electrons are required to constitute 1 coulomb of charge?

- Charge of 1 electron: $e = 1.6 \times 10^{-19}\ \mathrm{C}$

$$\text{Number of electrons} = \frac{1}{1.6 \times 10^{-19}} \approx 6.25 \times 10^{18}$$

4. Distinguish between conventional current and electron flow.

Aspect	Conventional Current	Electron Flow
Direction	Positive → Negative	Negative → Positive
Concept	Historical, assumed before electron discovery	Actual motion of electrons
Used in	Circuit diagrams, analysis	Physical electron motion explanation

5. What do you mean by the direction of electric current?

- **Definition:** The direction in which positive charges would move in a conductor.

- By convention, current is **from positive terminal to negative terminal**, even though electrons move opposite.

6. A charge of 96,500 C passes through a conductor in 1 hour. Calculate the current.

$$I = \frac{Q}{t}$$

$$Q = 96,500\,C, \quad t = 1\ \text{hour} = 3600\,s$$

$$I = \frac{96,500}{3600} \approx 26.8\,A$$

7. How much charge flows through a conductor when a current of 2 A flows for 10 minutes?

$$Q = I \cdot t$$

$$t = 10\ \text{min} = 600\,s$$

$$Q = 2 \times 600 = 1200\,C$$

8. Can the direction of current be opposite to the direction of electron flow? Explain.

- **Yes.** Conventional current is defined as positive → negative, while electrons move negative → positive. So **current direction is opposite to electron flow** in metals.

9. State the principle of conservation of charge.

- **Principle:** The total electric charge in an isolated system remains constant.
- Charge can **transfer** between bodies but **cannot be created or destroyed**.

10. What is the physical meaning of 1 ampere?

- **Definition:** 1 ampere is the current when **1 coulomb of charge flows per second**.

$$1\,A = 1\,\frac{C}{s}$$

11. Distinguish between DC and AC current.

Feature	DC (Direct Current)	AC (Alternating Current)
Direction	Constant, one way	Reverses periodically
Example	Battery, solar cell	Mains electricity
Symbol	I	$I(t) = I_0 \sin \omega t$
Uses	Electronics, charging	Household, transmission

12. In a conductor, how are electrons arranged?

- Electrons in a conductor have **two types of motion**:
    1. **Bound electrons:** attached to atoms (do not move freely).
    2. **Free electrons (conduction electrons):** move randomly and can drift under electric field.

13. What is the effect of temperature on the motion of free electrons in a conductor?

- **As temperature increases**, metal atoms vibrate more → **resistance increases**, **drift velocity decreases**.
- Conductivity decreases with temperature in metals.

14. Explain the concept of current density.

- **Definition:** Current density (J) is current per unit cross-sectional area.

$$J = \frac{I}{A}$$

- J is a vector in the direction of current flow.
- Unit: A/m^2

15. How does the current density relate to the electric field?

$$J = \sigma E$$

- σ = conductivity of the material.

- Stronger electric field → higher current density.

16. What is the relationship between current and number density of free electrons?

$$I = neAv_d$$

- n = number density of electrons
- e = charge of electron
- A = cross-sectional area
- v_d = drift velocity

17. A wire carries a current of 5 A. How many electrons pass through a cross-section in 2 seconds?

$$Q = I \cdot t = 5 \times 2 = 10\,C$$

$$\text{Number of electrons} = \frac{Q}{e} = \frac{10}{1.6 \times 10^{-19}} \approx 6.25 \times 10^{19}$$

18. What is meant by steady-state current?

- **Definition:** A current whose magnitude and direction **remain constant with time** in a circuit.
- Example: DC in a battery-operated circuit.

19. How does the current vary with time in a DC circuit?

- **In an ideal DC circuit**, current is constant over time ($I = \text{constant}$).
- Real circuits may have small transient variations initially.

20. Explain why current is considered to flow from positive to negative terminal.

- Historical convention: **Conventional current** assumes positive charges move.
- Even though electrons move from negative → positive, circuits are analyzed with **current flowing from positive to negative** for simplicity.

✅ SECTION 1 complete (Q1–20)

If you like, I can continue immediately with **SECTION 2: DRIFT VELOCITY & MOBILITY (Q21–45)** with the same **detailed explanations, formulas, and examples**.

Do you want me to continue to Section 2?

yes

Perfect! Let's move on to **SECTION 2: DRIFT VELOCITY & MOBILITY (Q21–45)** with detailed explanations, formulas, and examples.

SECTION 2: DRIFT VELOCITY & MOBILITY (Q21–45)

21. Define drift velocity.

- **Definition:** Drift velocity (v_d) is the **average velocity acquired by charge carriers (electrons) in a conductor due to an applied electric field**.
- Formula:

$$v_d = \frac{I}{neA}$$

Where:

- I = current,
- n = number density of electrons,
- e = electron charge,
- A = cross-sectional area.

22. Distinguish between random motion and drift velocity of electrons.

Feature	Random Motion	Drift Velocity
Cause	Thermal energy	Electric field

Feature	Random Motion	Drift Velocity
Direction	Random	Along field (opposite for electrons)
Speed	Very high (~10^5 m/s)	Very small (~10^{-4} m/s)
Contribution to current	Zero (cancels out)	Produces net current

23. What is the typical value of drift velocity in a conductor?

- Very small: $\sim 10^{-4}$ to 10^{-3} m/s under normal currents.
- Despite small drift velocity, the **effect of current is felt almost instantaneously** because the electric field propagates at near light speed.

24. Derive the relationship $I = neAv_d$.

- Consider a conductor of **cross-sectional area** A, **length** L, carrying **current** I.
- In time t, electrons move distance $v_d t$, so **volume of electrons passing**:

$$V = Av_d t$$

- Number of electrons: $N = n \cdot V = nAv_d t$
- Charge passing: $Q = Ne = neAv_d t$
- Current: $I = \frac{Q}{t} = neAv_d$ ✅

25. Define mobility of charge carriers.

- **Definition:** Mobility (μ) is the **drift velocity per unit electric field**.

$$\mu = \frac{v_d}{E}$$

- Measures **ease of motion of carriers in a material.**

26. What is the SI unit of mobility?

- $\mu = \frac{v_d}{E}$
- Units: $\frac{m/s}{V/m} = m^2 V^{-1} s^{-1}$

27. State the relationship between drift velocity and electric field.

$$v_d = \mu E$$

- Proportional relationship: **larger electric field → higher drift velocity**.

28. How does mobility depend on temperature?

- In **metals**, mobility decreases with **increase in temperature** due to increased lattice vibrations → more collisions.
- In **semiconductors**, mobility also decreases, but effect is smaller.

29. Why is the drift velocity very small compared to thermal velocity?

- Thermal velocity ($v_{th} \sim 10^5\ m/s$) arises from **random motion**.
- Drift velocity is due to **tiny net displacement along electric field**, so $v_d \ll v_{th}$.

30. In a conductor, if the electric field is doubled, what happens to drift velocity?

$$v_d = \mu E$$

- Doubling E → v_d **doubles**.
- Linear relation in ohmic conductors.

31. Explain the relaxation time of an electron.

- **Definition:** Average time (τ) between successive collisions of an electron with atoms in a conductor.
- Important for **calculating drift velocity and conductivity**.

32. What is the relationship between relaxation time and mobility?

$$\mu = \frac{e\tau}{m}$$

- e = electron charge, m = mass of electron.
- Higher τ → fewer collisions → higher mobility.

33. In a copper wire, what factors affect the drift velocity of electrons?

- Current (I)
- Number density (n)
- Cross-sectional area (A)
- Mobility of electrons (μ)

$$v_d = \frac{I}{neA} = \mu E$$

34. How does drift velocity depend on the cross-sectional area of the conductor?

$$v_d = \frac{I}{neA}$$

- **Smaller area → higher drift velocity** (for the same current).
- **Larger area → smaller drift velocity**.

35. Calculate drift velocity if $n = 8.5 \times 10^{28} m^{-3}, I = 10A, A = 10^{-6} m^2$.

$$v_d = \frac{I}{neA} = \frac{10}{(8.5 \times 10^{28})(1.6 \times 10^{-19})(10^{-6})}$$

Stepwise:

- Denominator: $8.5 \times 10^{28} \times 1.6 \times 10^{-19} = 1.36 \times 10^{10}$
- Multiply by $10^{-6} \rightarrow 1.36 \times 10^4$
- $v_d = 10/1.36 \times 10^4 \approx 7.35 \times 10^{-4} m/s$

36. Why do we need to apply an external electric field to produce current?

- Without an electric field, electrons move randomly → **no net current**.
- Electric field gives electrons a **preferred direction** → **net flow** → **current**.

37. What is the relationship between current and electric field?

$$I = neAv_d = neA\mu E$$

- Current is **directly proportional to electric field**.

38. Explain why drift velocity increases with applied voltage.

- Voltage V across conductor → electric field $E = V/L$
- $v_d = \mu E = \mu V/L$
- **Higher voltage → stronger field → faster drift**.

39. How does the number density of free electrons affect current?

$$I = neAv_d$$

- Higher n → more carriers → higher current (for same drift velocity).

40. In two wires of same material but different cross-sections, which has greater drift velocity for the same current?

- Smaller area → greater drift velocity: $v_d = I/neA$

41. Define relaxation time and explain its significance.

- Already defined in Q31.
- Significance: Determines **how quickly electrons respond to an electric field** and affects **resistivity and mobility**.

42. How is mobility related to the mean free path of electrons?

$$\mu = \frac{e\tau}{m}, \quad \lambda = v_{th}\tau$$

- λ = mean free path, τ = relaxation time.
- Larger λ → fewer collisions → higher mobility.

43. Why is mobility of electrons greater in semiconductors than in metals?

- Fewer free electrons → less electron-electron scattering
- Lattice scattering dominates
- Overall, electrons **move more easily per unit field** in semiconductors.

44. What happens to mobility when impurities are introduced in a conductor?

- Impurities → more collisions → **mobility decreases**.

45. Explain the concept of mean free path of an electron.

- **Definition:** Average distance an electron travels **between two successive collisions**.
- Symbol: $\lambda = v_{th}\tau$
- Longer λ → better conductivity.

✅ **SECTION 2 complete (Q21–45)**

If you want, I can continue with **SECTION 3: RESISTANCE & RESISTIVITY (Q46–85)** next, and we can maintain the **same detailed, formula-based approach**.

Do you want me to continue to Section 3?

yes

Great! Let's continue with **SECTION 3: RESISTANCE & RESISTIVITY (Q46–85)**. I'll keep the answers detailed, with formulas, examples, and explanations.

SECTION 3: RESISTANCE & RESISTIVITY (Q46–85)

46. Define resistance. State its SI unit.

- **Definition:** Resistance (R) is the **property of a conductor that opposes the flow of electric current**.
- Formula: $R = \frac{V}{I}$
- **SI unit:** Ohm (Ω), where $1\ \Omega = 1\ \mathrm{V/A}$.

47. State Ohm's law. Is it a universal law? Explain.

- **Statement:** The current through a conductor is **directly proportional** to the potential difference across it, **provided physical conditions remain constant**.

$$V \propto I \quad \Rightarrow \quad V = IR$$

- **Not universal:** Only valid for **ohmic materials**. Some devices (diodes, transistors) are **non-ohmic**.

48. Define resistivity and state its SI unit.

- **Definition:** Resistivity (ρ) is the **intrinsic property of a material that quantifies its resistance** per unit length and cross-section.

$$R = \rho \frac{L}{A}$$

- **SI unit:** Ohm-meter (Ω·m)

49. Derive the relationship $R = \rho \frac{L}{A}$.

- Current density: $J = \frac{I}{A}$
- Ohm's law: $J = \sigma E$, $\sigma = 1/\rho$
- Voltage: $V = EL \rightarrow E = V/L$
- $I/A = (1/\rho)(V/L) \rightarrow V/I = \rho L/A$

$$R = \rho \frac{L}{A}$$

50. How does resistance depend on the length of a conductor?

$$R \propto L$$

- **Longer conductor** → **higher resistance**, because electrons travel farther → more collisions.

51. How does resistance depend on the cross-sectional area of a conductor?

$$R \propto \frac{1}{A}$$

- **Thicker wire → lower resistance**, because more electrons can pass simultaneously.

52. What is the difference between resistance and resistivity?

Feature	Resistance (R)	Resistivity (ρ)
Depends on	Material + geometry	Material only
Formula	$R = \rho L/A$	Property, independent of shape
Unit	Ω	Ω·m

53. Why does resistivity depend on material but resistance depends on both material and geometry?

- **Resistivity** (ρ) is intrinsic: depends on **atomic structure and free electron density**.
- **Resistance** (R) also depends on **length and cross-section**, which affect collisions.

54. How does the resistivity of metals vary with temperature?

- **Metals:** resistivity increases with temperature due to **more lattice vibrations → more collisions**.

$$\rho_T = \rho_0(1 + \alpha \Delta T)$$

- α = temperature coefficient of resistance.

55. How does the resistivity of semiconductors vary with temperature?

- **Semiconductors:** resistivity decreases with temperature → more free electrons are released.
- Opposite trend to metals.

56. Define temperature coefficient of resistance.

- **Definition:** Fractional change in resistance per degree rise in temperature.

$$\alpha = \frac{R_T - R_0}{R_0 \Delta T}$$

57. What is the significance of a negative temperature coefficient?

- **Negative α:** Resistance decreases with temperature.
- Observed in **semiconductors**.
- Useful in **thermistors and temperature sensors**.

58. A copper wire of length L and cross-section A has resistance R. What will be the resistance of a wire of the same material with length 2L and area 2A?

$$R = \rho \frac{L}{A}, \quad R' = \rho \frac{2L}{2A} = \rho \frac{L}{A} = R$$

- **Resistance remains the same.**

59. Compare the resistivity of copper and nichrome.

Material	Resistivity (Ω·m)	Notes
Copper	1.68×10^{-8}	Very low, good conductor
Nichrome	1.1×10^{-6}	Higher resistivity, used in heating elements

60. Why are some materials ohmic while others are non-ohmic?

- **Ohmic:** Current ∝ Voltage (linear V–I) → constant resistance.
- **Non-ohmic:** Current not proportional to voltage → resistance changes with voltage or temperature.

61. A wire is stretched to double its length. How is its resistance affected?

- New area: $A' = A/2$ (volume constant)

$$R' = \rho \frac{2L}{A/2} = 4\rho \frac{L}{A} = 4R$$

- **Resistance quadruples.**

62. Two wires of copper and iron of the same length and area are connected in series. Which wire becomes hotter?

- **Power dissipated:** $P = I^2R$
- Iron has higher resistance → **iron wire gets hotter**.

63. Calculate the resistance of a copper wire of length 100 m and area $1mm^2$ ($\rho = 1.7 \times 10^{-8}\Omega m$)

$$R = \rho \frac{L}{A} = 1.7 \times 10^{-8} \frac{100}{1 \times 10^{-6}} = 1.7\Omega$$

64. What is the effect of temperature on the resistivity of a superconductor?

- Below critical temperature: **resistivity → 0**
- No resistance, perfect conductor.

65. Explain the concept of conductivity and conductance.

- **Conductivity** (σ): ability of material to conduct current

$$\sigma = \frac{1}{\rho}$$

- **Conductance (G):** ability of a specific conductor

$$G = \frac{1}{R}$$

- Unit: Siemens (S)

66. How are conductivity and resistivity related?

$$\sigma = \frac{1}{\rho}, \quad \rho = \frac{1}{\sigma}$$

67. What is the SI unit of conductance?

- **Siemens (S)**, where $1S = 1/\Omega$

68. A conducting wire is cut into two equal parts. What is the ratio of their resistances?

- Resistance of half wire: $R' = \rho \frac{L/2}{A} = R/2$
- Ratio: $R : R' = 2 : 1$

69. Why does a thin wire have greater resistance than a thick wire of the same material and length?

- $R \propto 1/A$
- Smaller cross-section → fewer paths for electrons → higher resistance.

70. What happens to the resistance of a conductor when it is bent?

- **Negligible change** (resistance depends on length and area).
- Slight effect if **stretching occurs**.

71. Explain why the filament of an electric bulb becomes very hot but connecting wires do not.

- Filament: very small cross-section → **high resistance** → **more power dissipated**
- Connecting wires: large area → small resistance → low heating.

72. Define specific resistance or resistivity with examples.

- Already defined in Q48.
- Examples: Copper ($1.68 \times 10^{-8}\Omega m$), Nichrome ($1.1 \times 10^{-6}\Omega m$), Glass ($\sim 10^{10}\Omega m$)

73. How does resistivity depend on the atomic structure of a material?

- Number of free electrons, lattice structure, and scattering determine resistivity.
- Metals: low resistivity, many free electrons.
- Insulators: high resistivity, few free electrons.

74. What is meant by resistance in series and parallel?

- **Series:** connected end-to-end, same current.

- **Parallel:** connected across same voltage, current divides.

75. In a metal, as temperature increases, does resistivity increase or decrease?

- **Increases** (more lattice vibrations → more collisions)

76. In a semiconductor, as temperature increases, does resistivity increase or decrease?

- **Decreases** (more charge carriers generated)

77. Explain the microscopic origin of resistance.

- Caused by **collisions of electrons with atoms/ions** in a conductor → energy dissipated as **heat**.

78. What is the relationship between resistivity and relaxation time?

$$\rho = \frac{m}{ne^2\tau}$$

- Longer τ → fewer collisions → lower resistivity.

79. How does resistivity change when a conductor undergoes strain?

- Stretching: **resistivity increases** slightly (geometry changes, more collisions)

80. Why do we need to know the resistivity of materials?

- To **design circuits**, choose materials for conductors, insulators, heating elements.

81. In what way does Ohm's law fail in non-ohmic devices?

- $V \not\propto I$ → resistance varies with **voltage, temperature, or time**.
- Example: diode, filament bulb.

82. Calculate resistivity if $R = 10\Omega, L = 2m, A = 0.5mm^2$

$$\rho = R\frac{A}{L} = 10\frac{0.5 \times 10^{-6}}{2} = 2.5 \times 10^{-6}\Omega m$$

83. A metal has temperature coefficient 0.005/°C. Resistance at 0°C = 100 Ω. Find R at 100°C.

$$R = R_0(1 + \alpha\Delta T) = 100(1 + 0.005 \times 100) = 100(1 + 0.5) = 150\Omega$$

84. Why are alloys like nichrome used in heating elements?

- **High resistivity** → good heat generation
- **Low temperature coefficient** → stable resistance

85. What is the difference between resistivity of conductors, semiconductors, and insulators?

Material	Resistivity (Ω·m)
Conductor	$10^{-8} - 10^{-6}$
Semiconductor	$10^{-3} - 10^{3}$
Insulator	$10^{8} - 10^{20}$

✅ **SECTION 3 complete (Q46–85)**

I can now continue with **SECTION 4: EMF & INTERNAL RESISTANCE (Q86–110)** next.

Shall I go ahead?

yes

Perfect! Let’s move on to **SECTION 4: EMF & INTERNAL RESISTANCE (Q86–110)** with detailed explanations, formulas, and examples.

SECTION 4: EMF & INTERNAL RESISTANCE (Q86–110)

86. Define EMF (electromotive force).

- **Definition:** EMF (E) of a source is the **work done per unit charge** to move charge from the negative to positive terminal **inside the source**.

$$\mathrm{E} = \frac{W}{q}$$

87. What is the SI unit of EMF?

- **Volt (V)**, same as potential difference.

88. Distinguish between EMF and potential difference.

Feature	EMF (ε)	Potential Difference (V)
Location	Inside source	Across any two points of a circuit
Cause	Chemical or other energy	Work done by electric field
Current	Maximum when no load	Depends on current
Symbol	ε	V

89. What is internal resistance? Why does every battery have internal resistance?

- **Internal resistance (r):** resistance offered by **electrolyte and electrodes** inside a cell.
- Causes: ionic friction, chemical limitations.
- **Every battery has r** because no material is a perfect conductor.

90. State the relationship between terminal voltage, EMF, and internal resistance.

$$V = \mathrm{E} - Ir$$

- I = current, V = voltage across terminals.

91. When is the terminal voltage equal to EMF?

- **Open circuit** $(I = 0) \rightarrow V = \mathrm{E}$

92. When is the terminal voltage less than EMF?

- **When current flows** $(I > 0) \rightarrow V = \mathrm{E} - Ir < \mathrm{E}$

93. A battery of EMF 12 V and internal resistance 0.5 Ω is connected to a load of 11.5 Ω. Find current and terminal voltage.

- Total resistance: $R + r = 11.5 + 0.5 = 12\Omega$

$$I = \frac{\mathrm{E}}{R + r} = \frac{12}{12} = 1A$$

$$V = \mathrm{E} - Ir = 12 - 1 \cdot 0.5 = 11.5V$$

94. What happens to the terminal voltage when current drawn increases?

- $V = \mathrm{E} - Ir \rightarrow$ **terminal voltage decreases**.

95. Explain why a cell can supply only limited current.

- Due to **internal resistance**: $I_{max} = \frac{\mathrm{E}}{r}$
- Exceeding this → voltage drops, heating occurs.

96. What is the difference between an ideal and a real cell?

Feature	Ideal Cell	Real Cell
Internal resistance	0 Ω	Non-zero
Terminal voltage	Always = EMF	Decreases with current
Power loss	0	Present in r

97. How does internal resistance affect efficiency of a battery?

$$\eta = \frac{P_{load}}{P_{total}} = \frac{I^2R}{I^2(R+r)} = \frac{R}{R+r}$$

- Larger r → lower efficiency.

98. Derive $V = \mathrm{E} - Ir$.

- Total EMF: $\mathrm{E} = V + Ir$
- Terminal voltage: $V = \mathrm{E} - Ir$

99. When is power dissipated in internal resistance maximum?

- Power in r: $P_r = I^2r$
- For maximum power delivered to load: $R = r$
- Then $P_{load} = P_r = \frac{\mathrm{E}^2}{4r}$

100. How can internal resistance of a battery be determined experimentally?

- Connect variable load, measure **current (I)** and **terminal voltage (V)**.
- Plot V vs I → slope = $-r$, intercept = EMF (E).

101. What is meant by the loading effect of internal resistance?

- When the load draws current, **terminal voltage drops** due to internal resistance → **affects device performance**.

102. Explain why maximum power is delivered when $R = r$.

- Power to load: $P = I^2R = \frac{\mathrm{E}^2R}{(R+r)^2}$
- Differentiate w.r.t R, set $\frac{dP}{dR} = 0 \rightarrow R = r$

103. Why does a battery become warm during discharge?

- Due to **power dissipated in internal resistance**: $P = I^2r$

104. What happens to terminal voltage while charging a battery?

- Current opposite to discharge → voltage **greater than EMF**: $V = \mathrm{E} + Ir$

105. Explain back EMF in a charging cell.

- During charging, battery opposes applied voltage → **back EMF** generated inside cell.

106. Can internal resistance of a battery be zero? Explain.

- No. **Every real battery has some internal resistance** due to electrolyte and electrodes.

107. How does temperature affect internal resistance?

- Higher temperature → decreased resistance in metals
- Lower temperature → increased resistance
- Electrolyte conductivity also changes.

108. In a circuit with multiple batteries, how is net EMF calculated?

- **Series:** Sum EMFs (consider polarity)
- **Parallel:** Same voltage (if ideal), otherwise depends on internal resistance.

109. How does EMF of a cell differ during charging and discharging?

- **Discharging:** EMF = chemical energy → decreases slightly due to r.
- **Charging:** EMF + applied voltage → cell gains energy.

110. Why does terminal voltage decrease when more devices are connected in parallel?

- More current drawn → $V = \mathrm{E} - Ir$ → terminal voltage drops due to internal resistance.

✅ **SECTION 4 complete (Q86–110)**

Next, we can cover **SECTION 5: OHM'S LAW & KIRCHHOFF'S LAWS (Q111–130)** in the same detailed format.

Shall I continue?

yes

Perfect! Let's continue with **SECTION 5: OHM'S LAW & KIRCHHOFF'S LAWS (Q111–130)**, providing detailed explanations, formulas, and examples.

SECTION 5: OHM'S LAW & KIRCHHOFF'S LAWS (Q111–130)

111. State Ohm's law (macroscopic form).

- **Statement:** The potential difference V across a conductor is **directly proportional** to the current I passing through it, provided temperature and other physical conditions remain constant.

$$V = IR$$

112. State microscopic form of Ohm's law.

- Relates **current density** J to **electric field** E:

$$\mathbf{J} = \sigma \mathbf{E}$$

- σ = conductivity of the material.

113. What is the condition for validity of Ohm's law?

- Material must be **ohmic**
- Temperature, pressure, and other conditions **must remain constant**
- Linear V–I relationship

114. What are ohmic and non-ohmic devices? Give examples.

Type	Definition	Example
Ohmic	V ∝ I (linear)	Resistor, metallic wire
Non-ohmic	V ∝̸ I (nonlinear)	Diode, filament bulb

115. Draw a V–I graph for an ohmic resistor.

- **Graph:** Straight line passing through origin, slope = $1/R$

116. Draw a V–I graph for a non-ohmic device (e.g., diode).

- **Graph:** Nonlinear curve, current rises sharply after threshold voltage.

117. State Kirchhoff's current law (KCL).

- **Law:** The algebraic sum of currents at any junction is zero.

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

- **Based on:** conservation of charge.

118. State Kirchhoff's voltage law (KVL).

- **Law:** In any closed loop, the algebraic sum of potential differences equals zero.

$$\sum V = 0$$

- **Based on:** conservation of energy.

119. What physical principle underlies Kirchhoff's current law?

- **Conservation of charge**: charge cannot accumulate at a junction.

120. What physical principle underlies Kirchhoff's voltage law?

- **Conservation of energy**: energy gained = energy lost around a loop.

121. In a series circuit: V_1 = 2 V, V_2 = 3 V, total EMF = 6 V. Find V_3.

- KVL: $V_{\text{total}} = V_1 + V_2 + V_3$

$$6 = 2 + 3 + V_3 \quad \Rightarrow \quad V_3 = 1 \text{ V}$$

122. Use Kirchhoff's laws to find current in each branch of a network.

- **Method:**
  1. Assign current directions.
  2. Apply KCL at junctions.
  3. Apply KVL in loops.
  4. Solve simultaneous equations.

123. How many independent equations can be written in a circuit with n loops?

- **Number of KVL equations:** n (number of loops)
- **Number of KCL equations:** (number of junctions - 1)

124. Can Kirchhoff's laws be applied to AC circuits?

- **Yes**, provided voltages and currents are **instantaneous or phasor quantities**.

125. Can sum of potential rises equal sum of potential drops in a loop? Explain.

- **Yes**, KVL states: sum of EMFs (rises) = sum of voltage drops.

126. What is the significance of Kirchhoff's laws?

- Solve **complex circuits**
- Predict **current and voltage in each branch**
- Apply **conservation laws**

127. How can Kirchhoff's laws be used to find equivalent resistance?

- Replace parts of the circuit with **unknown resistance**, use **KCL and KVL** to derive total current and voltage → calculate R_{eq}.

128. Explain why Kirchhoff's laws are based on conservation laws.

- **KCL:** conservation of charge (current into a junction = current out)
- **KVL:** conservation of energy (sum of potential differences = 0)

129. How do we choose current direction in multi-loop circuits?

- Arbitrary choice; **if solution is negative**, actual current flows opposite to assumed direction.

130. Can Kirchhoff's voltage law be applied when magnetic field is time-varying? Explain.

- **No**, Faraday's law introduces induced EMF:

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt} \neq 0$$

- KVL fails for **time-varying magnetic fields**.

✅ **SECTION 5 complete (Q111–130)**

Next, we can move on to **SECTION 6: SERIES & PARALLEL COMBINATIONS (Q131–150)**.

Shall I continue with that?

yes

Perfect! Let's complete the last section: **SECTION 6: SERIES & PARALLEL COMBINATIONS (Q131–150)** with detailed explanations, formulas, and examples.

SECTION 6: SERIES & PARALLEL COMBINATIONS (Q131–150)

131. What is meant by resistors in series? Draw a diagram.

- **Definition:** Resistors are in **series** if connected end-to-end so that **same current** flows through each.
- **Diagram:**

```
---R1---R2---R3---
```

132. Derive formula for equivalent resistance in series.

- Current same through all: I
- Total voltage: $V = V_1 + V_2 + V_3 = I(R_1 + R_2 + R_3)$

$$R_{\text{eq}} = R_1 + R_2 + R_3$$

133. In series combination, does current remain same?

- **Yes**, same current flows through all resistors.

134. In series combination, does voltage vary?

- **Yes**, voltage divides among resistors:

$$V_i = IR_i$$

135. What is the advantage of series combination?

- Simple, **voltage division** possible, useful for creating required voltage drops.

136. What is the disadvantage of series combination?

- If one resistor fails → **entire circuit breaks**.

137. What are resistors in parallel? Draw a diagram.

- **Definition:** Resistors are in **parallel** if connected across same two points, sharing **same voltage**.
- **Diagram:**

```
    |---R1---|
 ---|        |---
    |---R2---|
    |---R3---|
```

138. Derive formula for equivalent resistance in parallel.

- Voltage same: V
- Total current: $I = I_1 + I_2 + I_3 = V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

139. In parallel combination, does current remain same?

- **No**, current divides: $I = I_1 + I_2 + I_3$

140. In parallel combination, does voltage remain same?

- **Yes**, same across all resistors.

141. What is the advantage of parallel combination?

- Voltage uniform across devices, failure of one branch **does not affect others**.

142. What is the disadvantage of parallel combination?

- More wires needed, complex arrangement.

143. Three resistors 2 Ω, 3 Ω, 6 Ω are in parallel. Find their equivalent resistance.

$$\frac{1}{R_{\text{eq}}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1 \quad \Rightarrow \quad R_{\text{eq}} = 1\,\Omega$$

144. Three resistors 2 Ω, 3 Ω, 6 Ω are in series. Find total resistance.

$$R_{\text{eq}} = 2 + 3 + 6 = 11\,\Omega$$

145. Explain current divider rule with derivation.

- **Parallel resistors** R_1 and R_2, total current I splits:

$$I_1 = I\frac{R_2}{R_1 + R_2}, \quad I_2 = I\frac{R_1}{R_1 + R_2}$$

- Current **inversely proportional** to resistance.

146. Explain voltage divider rule with derivation.

- **Series resistors:** R_1, R_2, total voltage V

$$V_1 = V\frac{R_1}{R_1 + R_2}, \quad V_2 = V\frac{R_2}{R_1 + R_2}$$

- Voltage **proportional to resistance** in series.

147. In a parallel circuit, if one resistor is zero, what happens?

- Acts as **short circuit**, total resistance → 0, current → very high (dangerous).

148. If a series resistor becomes very large, what happens to current?

- Total resistance ↑ → current ↓ (Ohm's law: $I = V/R$)

149. Can series and parallel resistors exist in one circuit?

- **Yes**, most real circuits are **mixed** (series-parallel combination).

150. How is total resistance calculated in a mixed combination?

1. Combine **series resistors** first → replace with $R_{\text{eq,series}}$
2. Combine **parallel resistors** → replace with $R_{\text{eq,parallel}}$
3. Repeat until **single equivalent resistance** obtained.